



Full Stack Developer Technical Assessment

Position: Senior Laravel + Vue Developer

Overview

Build a **Dynamic Commission Engine** that allows administrators to create and manage commission formulas without changing application code.

The solution should demonstrate backend architecture, frontend design, problem-solving ability, and scalability considerations.

Business Scenario

Energy brokers receive commission based on customer contracts.

Commission calculations must be configurable by administrators and support multiple versions over time.

Example:

Version 1

$\text{Commission} = \text{Annual Usage} \times 0.05$

Version 2

$\text{Commission} = (\text{Annual Usage} \times 0.05) + (\text{Contract Length} \times 100)$

The system should support creating, testing, and activating new commission formulas.

Functional Requirements

1. Formula Builder

Create a UI that allows administrators to create commission formulas.

Supported variables:

- Annual Usage
- Contract Value
- Contract Length
- Risk Score

Example formula:

$(\text{AnnualUsage} * 0.05) + (\text{ContractLength} * 100)$

Requirements:

- Save formulas in the database
- Validate formulas before saving
- Support multiple versions
- Only one version can be active at a time

2. Commission Calculation

Given a contract, calculate the commission using the active formula.

Display:

- Formula Version
- Input Values
- Calculation Result

Store calculation history for future reference.

3. Formula Impact Analysis

Before activating a new formula, provide a simulation mode.

The simulation should show:

- Number of affected contracts
- Current total commission
- New total commission
- Difference between the two

The simulation must not modify existing commission records.

4. Dependency Validation

Support calculated variables.

Example:

$\text{BaseCommission} = \text{AnnualUsage} * 0.05$

$\text{BonusCommission} = \text{BaseCommission} * 0.10$

$\text{FinalCommission} = \text{BaseCommission} + \text{BonusCommission}$

Requirements:

- Detect circular dependencies
- Prevent invalid formulas
- Calculate in the correct sequence

5. Audit Trail

Every commission calculation should be traceable.

Display:

- Formula Version
- Calculation Date
- Input Values
- Calculation Steps
- Final Result

Technical Requirements

Backend

- Laravel
- REST API
- Queue Jobs where appropriate
- Unit Tests

Frontend

- Vue

Database

- MySQL

Deliverables

- Source Code
- README
- Database Schema

- API Documentation
- Test Cases
- Architecture Notes

The focus is on maintainable architecture and clean implementation rather than UI styling.